B.Tech. Degree V Semester Second Internal Examination November 2024

Department of Mechanical Engineering

19-200-0501 NUMERICAL AND STATISTICAL METHODS

Time: 2 Hours　　　　　　　　　　　　　　　　　　　　Maximum Marks: 40

CO3: Compute the mean and variance of a probability distribution including the binomial distribution.

CO4: Test hypotheses on data.

PART A　　　　　　　　　　　　　　(4x4=16)

I		Marks	CO	BL
a ✓	A random variable X has the following probability distribution. X: -2, -1, 0, 1, 2, 3 P(X=x): $\frac{1}{10}$, k, $\frac{1}{5}$, 2k, $\frac{3}{10}$, 3k Find (i) the value of k (ii) P(-2 < X < 2) (iii) P(X < 2)	4	3	L2
b ✓	Determine the probability that there are 3 defective items in a sample of 100 items if 2% of items made in this factory are defective.	4	3	L3
c ✓	Define (i) Parameter (ii) Type I error (iii) Alternative hypothesis (iv) Population.	4	4	L1
d ✓	Determine the probability that the sample mean will be between 75 and 78 if a random sample of size 100 is taken from a population having mean 76 and variance 256.	4	4	L2

PART B　　　　　　　　　　　　　　(12x2=24)

		Marks	CO	BL
II a✓	Derive mean and variance of binomial distribution.	6	3	L1
b✓	Fit a least square straight line to the following data: X: 2, 7, 9, 1, 5, 12 Y: 13, 21, 23, 14, 15, 21	6	3	L2
	OR			
III a	The probability of a man hitting a target is $\frac{1}{3}$. If he fires 5 times, what is the probability of his hitting the target (i) at least twice (ii) at most 3 (iii) exactly 4.	6	3	L3
b	Determine the expected number of boys whose weight is (i) between 65 and 70 kg (ii) greater than 72 kg (iii) less than 75 kg, if the weight of 800 boys follows normal distribution with mean 66 and standard deviation 5.	6	3	L3
IV a✓	A sample of 900 numbers is found to have a mean 3.4 and standard deviation 2.61. Is this sample taken from a large population of mean 3.25?	6	4	L3

✓b	The standard deviation of a sample of 20 observations from a normal population was found to be 5. Examine whether the sample was taken from a population with standard deviation 5.3.	6	4	L3
	OR			
V a	In a random sample of 100 tube lights produced by a company A, the mean life time of tube light is 1190 hrs with standard deviation of 90 hrs. Also in a random sample of 75 tube lights from company B the mean lifetime is 1230 hrs with standard deviation 120 hrs. Is there a difference between the mean lifetime of the two brands of the tube lights?	6	4	L3
b	The intelligence test of 2 groups of boys and girls gives the following results Girls — S.D = 10 — n = 121 Boys — S.D = 12 — n = 81 Is the difference in S.Ds significant?	6	4	L3

Girls	S.D = 10	n = 121
Boys	S.D = 12	n = 81

SF3

EVEN NOS

School of Engineering

Cochin University of Science And Technology

B.Tech Degree V[th] Semester first series test in Mechanical Engineering (A&B)

Time: 2Hrs **Sub: 19-205-0505:POWER PLANT ENGINEERING** Max.Marks:25

(Answer all questions , ***Answer Section A and B in separate answer books)***

Q. No	Section A			
1 (a)	At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as follows : Transformer 1 General power service and lighting: a : 10 H.P., 5kW; b : 7·5 H.P., 4kW; c : 15 H.P.; d : 5 H.P., 2 kW Transformer 2 Residence lighting: e : 5 kW; f : 4 kW; g : 8 kW; h : 15 kW; i : 20 kW Transformer 3 Store lighting and power: j : 10 kW, 5 H.P.; k : 8 kW, 25 H.P.; l : 4 kW The H.P. load is motor load and assume an efficiency of 72%. The diversity factor among consumers of Transformers 1,2 &3 is 1.5 and diversity factor between transformers 1,2&3 is 1.3.Assume that the power factors of consumers a & b to be 70% and that of c,d,j&k to be 75% and predict the maximum demand on the feeder. (10 Marks)			
(b)	The capital cost of a power generating equipment in a steam power plant is Rs 80 X10^6. The useful life of the plant is 30 years and its salvage value is 5% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6%. (10 Marks)			
	Section B(Answer in a separate answer sheet)			
2	Explain different components of a thermal power plant. Draw the T-s and h-S diagram for a simple cycle and explain each systems of the plant against the diagram (10 Marks)			

SCHOOL OF ENGINEERING, CUSAT
DIVISION OF MECHANICAL ENGINEERING
V SEMESTER INTERNAL EXAM II

19-205-0502 MECHANICS OF MACHINERY (SEM V-B)

Time : 2hours **Max. Marks 40**

PART A **(2X5=10)**

Q1. Why prime circle is used while drawing certain cam profiles?

Q2. What is the difference between radial and offset follower?

Q3. What is meant by displacement diagram related to cam kinematics.

Q4. Derive velocity ratio for a Gear train which include a compound gear.

Q5. What do you mean by epicyclic gear-train? Where is it used?

PART B **(15X2=30)**

Q1. A disc cam used for moving a roller-end follower with simple harmonic motion during lift and uniform acceleration and retardation motion during return rotates in clockwise direction at 300 rpm. The line of motion of the follower has an offset 10 mm to the right of camshaft axis. The minimum radius of the cam is 30 mm. The lift of the follower is 40 mm. The cam rotation angles are: Lift 60° dwell 90° return 120° and remaining angle for dwell. Draw the cam profile and determine the maximum velocity and acceleration during the lift and return.

Q2. In an epicyclic gear train, the internal wheels A and B and compound wheels C and D rotate independently about axis O. The wheels E and F rotate on pins fixed to the arm G. E gears with A and C. Wheel F gears with B and D. All the wheels have the same module and the number of the teeth are: T_C=35; T_D= 30; T_E= T_F= 20.

1. Sketch the arrangement.
2. Find the number of teeth on A and B.
3. If the arm G makes 125 rpm clockwise and A is fixed, find the speed of B.
4. If the arm G makes 125 rpm clockwise and wheel A makes 15 rpm anticlockwise; find the speed of wheel B.

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B. TECH DEGREE 5TH SEMESTER SECOND INTERNAL EXAMINATIONS,

NOVEMBER -2024

19-205-0605 (IE): INDUSTRIAL MANAGEMENT

MECHANICAL ENGINEERING

TIME: 2 Hrs **MAX. MARKS: 40**

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the concept of management, organization and personal management and apply those concepts in industrial environment
CO2	Get the concept of industrial relations and use it in real life
CO3	Understand the concept of financial management, project management and economics to use in an industrial situation.
CO4	Study the concept of marketing management and basics of IPR.

PART A

(Answer all questions) **(4 x 4= 16 Marks)**

Q.No	Questions	Marks	BL	CO	PO
I (a)	Differentiate between wages and incentives.	4	L1	CO2	PO1
(b)	Discuss the importance of manager inventory charts.	4	L2	CO2	PO1
(c)	Differentiate between consumer markets and industrial markets.	4	L2	CO4	PO1
(d)	What are the highlights of the 1970 Indian Patent act?	4	L1	CO4	PO1

PART B

(Answer the following questions) **(2 x 12= 24 Marks)**

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Explain the system approach of selection.	6	L2	CO2	PO1, PO2
(b)	Discuss the various training methods followed in an industry	6	L2	CO2	PO1, PO2
	OR				
III (a)	Discuss the methods used to find the relative worth of an employee and a job with respect to an organization.	6	L2	CO2	PO1
(b)	Discuss the reasons for industrial accidents in an organization.	6	L2	CO2	PO1, PO2

IV (a)	What is the purpose of market segmentation? Explain different parameters by which markets are segmented.	6	L2	CO4	PO1
(b)	Explain different methods which are used to do market research.	6	L3	CO4	PO1
	OR				
V (a)	Describe various forms of IPR citing their significance.	6	L2	CO4	PO1,PO2
V (b)	Explain the steps involved to file a patent in India. What are the eligibility criteria for obtaining a patent?	6	L1	CO4	PO1,PO2

L1=30%, L2=60%, L3=10%, L4=0%, L5=0%, L6=0%

School Of Engineering CUSAT
Second Internal Examination-Nov2024
19-205-0504 Thermal Engineering

Time: 2 Hrs **(Answer all Questions)** **Max marks 30**

PART A

		Mark	BL	CO	PO
1.	Classify fuels giving examples for each.	3	2	1	1
2	Explain Adiabatic Flame Temperature.	3	2	1	1
3	Explain SI engine and CI engine fuel rating	3	2	5	1
4	Explain supercharging of CI engine and its effect on indicator diagram.	3	2	4	1

PART B

		Mark	BL	CO	PO
5	a)Explain application of first law of thermodynamics to steady flow combustion	4	2	2	1
	b)Liquid Propane (C_3H_8) enters a combustion chamber at 25^{0}C at a rate of 0.05kg/min where it is mixed and burned with 50% excess air that enters the combustion chamber at 7^{0}C. An analysis of the combustion gases reveals that all the hydrogen in the fuel burns to H_2O but only 90% of the carbon burns to CO_2, with the remaining 10% forming CO. If the exit temperature of the combustion gases is 1500K, determine the mass flow rate of air and the rate of heat transfer from the chamber.	8			
6	a)Explain with figures, phases of normal combustion in CI engines	4	2	5	1
	b)Explain with figures, normal combustion, Pre-ignition and detonation in SI engines.	4		5	
	c)Explain with figures, Combustion chamber shapes of CI engines	4		4	

FIFTH SEMESTER B.TECH. (MECHANICAL ENGG)
SECOND INTERNAL EXAMINATION, NOV 2024
19-205-0503 Machining Science and Machine Tools
(B batch)

Time: 2 Hours
Marks: 50

CO1	To study the geometry of single point and multipoint cutting tools and m chip formation
CO2	To get the concept of mechanics of metal cutting
CO3	To understand the principles of working, specifications, different types ar performed in lathe, drilling and shaping machines
CO4	To learn the principles of working, specifications, different types and ope performed in grinding and milling machines

Part A
Answer all questions

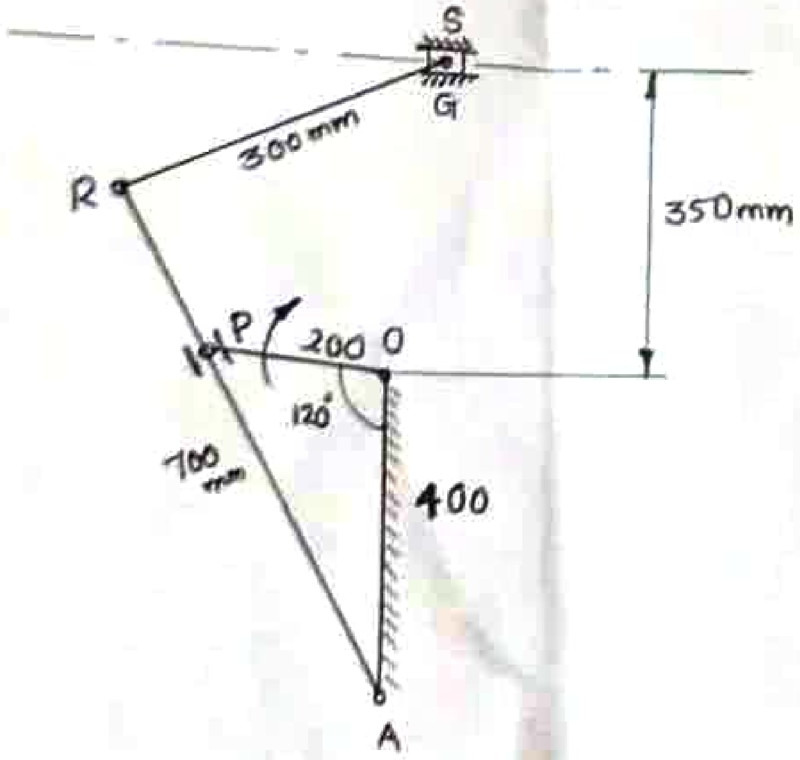
I		Marks	BL	CO	PO
a	How does the shear angle afffect the cutting process ?	3			
b	Explain the significance of the cutting force and thrust force in Orthogonal machining?	3			
c	Explain the mechanisms of crator wear and flank wear in cutting tools.	3			
d	What factors influence the rate of tool wear in metal cutting?	3			
e	How does the tool reciprocates in a shaper?	3			
f	What are the limitations of a shaper compared to other machine tools?	3			

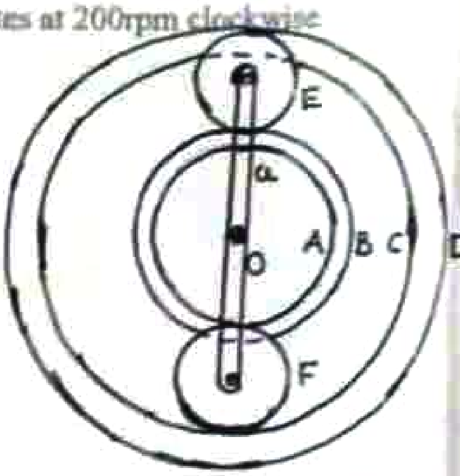
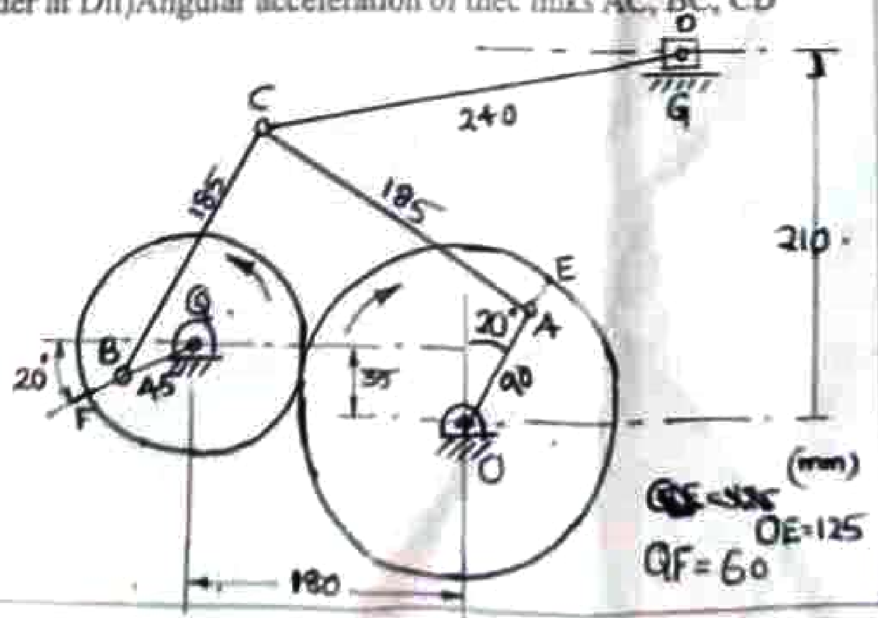
PART B

II		Marks	BL	CO	PO
	(a) A tool with a rake angle of 10 degrees cuts a metal with a shear angle of 25 degrees. If the friction angle between the tool and the chip is 30	6			

	degrees, determine the shear force given that the cutting force is 800 N. (b) The cutting speed of a machining operation is 120 m/min and the rake angle of the tool is 15 degrees. If the chip thickness ratio is 0.6, calculate the shear angle.	6			
III	(a) A cylindrical workpiece with a dia of 60 mm is machined in a lathe with a feed rate of 0.2 mm/rev and a depth of cut of 1 mm. Calculate the material removal rate if the spindle speed is 500 rpm. (b) Calculate the machining time required to turn a 150 mm long workpiece from a dia of 100 mm to 90 mm, given a spindle speed of 400 rpm and a feed rate of 0.25 mm/rev.	6 6			
IV	(a) Explain the quick return mechanism in a shaping machine with suitable sketeches. (b) What are the main types of tool wear observed in metal cutting? Describe each type and its effect on the cutting process,	6 6			

Time: 2hrs	Department of Mechanical Engineering, SOE Second Internal Examination of S-5 B.Tech Mechanical Engineering November 2024 19-205-0502 MECHANICS OF MACHINERY	Max Marks:40

Q No	ANSWER ANY FOUR QUESTIONS (4X10=40)	CO	BL	Marks
1.	i)Define transmission angle? ii) A crank-rocker mechanism has a 70mm fixed link,a 20mm crank,a 50mm coupler,and a70mm rocker.Draw the mechanism and determine the maximum and minimum values of the transmission angle. Locate the two toggle positions and find the corresponding crank angles and the transmission angles	CO2	L2	10
2.	i)- Figure shows the link mechanism of a quick return mechanism of the slotted lever type,the various dimensions of which are OA=400mm,OP=200mm,AR=700mm,RS=300mm.For the configuration shown determine the acceleration of the cutting tool at S and the angular acceleration of the link RS.The crank OP rotates at210rpm 	CO1	L3	10

3.	i)Define i)Prime circle of a cam ii)Pressure angle of a cam ii)Draw the profile of a cam operating a roller reciprocating follower and with minimum radius of the cam=25mm,Lift=30mm,Roller diameter=15mm The cam lifts the follower for 120° with cycloidal motion followed by a dwell period of 30°.Then the follower lowers down during 150° of the cam rotation with uniform acceleration and deceleration followed by a dwell period.If the cam rotates at a uniform speed of 150rpm ,Calculate the maximum velocity and acceleration of the follower during the descent period	CO2	L3	10
4.	i)-Define i).Velocity ratio(Gear ratio)ii)Module ii)-In the epicyclic gear train shown in the figure the compound wheels A and B as well as internal wheels C and D rotate independently about the axis O.Th e wheels E and F rotate on the pins fixed to the arm a.All the wheels are of the same module.The number of teeth on the wheels are T_A=52,T_B=56,T_E=T_F=36 Determine the speed of C if i)The wheel D fixed and arm a rotates at 200rpm clockwise ii)The wheel D rotates at 20rpm counter clockwise and trhe arm a rotates at 200rpm clockwise  	CO2	L3	10
5.	An Andrew variable stroke engine mechanism is shown ifn fif .The crank OA ritates at 100rpm.Find the i) linear acceleration of slider at Dii)Angular acceleration of thec links AC, BC, CD  	CO2	L2	10

School of Engineering

Cochin University of Science And Technology

B.Tech Degree Vth Semester second series test in Mechanical Engineering (A batch)

Time:2Hrs **Sub:19-205-0504 Thermal Engineering** Max.Marks: 50

Faculty:Dr. Gireesh Thampi (Answer all questions. Appropriate charts are permitted)

CO1 To understand the combustion phenomenon in IC engines
CO3 To apply the knowledge of thermodynamics of combustion
CO4 To analyse the performance of IC engines

Q.No		CO	L	PO
I (a)	(i) Explain the following terms (i) Stoichiometric combustion (ii) Equivalence ratio (iii) Ultimate and proximate analysis. (4) (ii)Determine the gravimetric analysis of the products of combustion of acetylene with 200% stoichiometric air. (5) (iii)One kg of octane is burned with 200% theoretical air. Assuming complete combustion, determine (i) A:F (ii) Dew point of the products at a pressure of 100kPa. (6)	1 3	3	1,12 3,12 3, 12
I (b)	(i) Explain the following terms (i) Enthalpy of formation (ii) Enthalpy of combustion (iii) Enthalpy of reaction. (3) (ii)A gasoline engine delivers 150kW. The fuel used is Octane and the air enters at 4-^{0}C. The products of combustion leave the engine at750K, and the heat transfer from the engine is 205kW. Determine the fuel consumption per hour if complete combustion is achieved. (8)	1 3	3	3, 12
II(a)	A test on a single,4stroke oil engine having bore 180mm and stroke 360 mm gave the following results: speed-290rpm, brake torque-392Nm, indicated mean effective pressure 7.2bar, oil consumption- 3.5kg/Hr, cooling water flow- 270kg/Hr, cooling water temperature rise-36^{0}C, A:F ratio by weight-25, exhaust gas temperature- 415^{0}C, barometric pressure-1.013bar,room temperature-21^{0}C .The fuel has a calorific value of 45200kJ/kg and contains 15 %of hydrogen by weight. Calculate (a) the indicated thermal efficiency (ii) volumetric efficiency based on atmospheric conditions (iii) Draw up a heat balance sheet interms of kJ/min . Take R= 0.287kJ/kgK, C_p for dry exhaust gases =1.0035kJ/kgK and C_p for superheated steam = 2.093kJ/kgK. (10)	4	4	3,12
II(b)	Explain a suitable method to determine the indicated power of an engine for a constant fuel flow rate. (3)	3	3	2,12
III	(i) Describe the stages of combustion in SI engines (4) (ii) Explain the phenomenon of auto ignition. How auto ignition influence knocking. (4) (iii) What is HUCR. How it is determined and what is its significance? (3)	1	2	2,12

xxxxxx